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PG PROJECT REPORT

OncoDet: An AI-Based Lung Cancer Detection System Using Chest X-Ray Radiographs

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ABSTRACT

Lung cancer is among the leading causes of cancer-related death both globally and within Central Asia. In Uzbekistan, the combination of rising cancer incidence, a shortage of qualified radiologists, and limited diagnostic infrastructure outside major urban centres means that many patients reach specialist oncology services only when their disease has already progressed to an advanced stage. This project introduces OncoDet, a deep learning-based clinical decision-support system designed to assist physicians in identifying lung cancer-related pathology from chest radiograph images.

The system uses DenseNet121, a densely connected convolutional neural network, pre-trained on the ImageNet large-scale visual recognition dataset and subsequently fine-tuned on a collection of 128 de-identified chest X-ray images assembled from the Republican Specialized Scientific-Practical Medical Center of Oncology and Radiology (RSSPMCOiR) in Tashkent. Training used Focal Loss to address the imbalance between cancer-positive and normal cases that is inherent in clinical data. The best model checkpoint achieved a validation accuracy of 86.7% and an F1-score of 0.833 on a balanced held-out validation set.

Beyond classification, OncoDet incorporates three supporting modules that are central to clinical usability: a Gradient-weighted Class Activation Mapping (Grad-CAM) component that overlays a colour heatmap on the original X-ray to show which regions influenced the prediction; a multi-check out-of-distribution detector that rejects non-radiograph uploads before inference; and a federated learning simulation based on the Federated Averaging algorithm (FedAvg) that demonstrates how the model could be trained across multiple hospital nodes without sharing patient data. The complete system is packaged as a full-stack web application, with a Flask REST API serving the model and a React frontend providing the clinical interface.

This work addresses a documented gap in Uzbekistan's oncological diagnostic capability and demonstrates that state-of-the-art deep learning can be applied meaningfully even to locally collected clinical datasets that are modest in size.

Keywords: *lung cancer detection, DenseNet121, deep learning, transfer learning, Grad-CAM, Uzbekistan clinical data,*

1. Introduction

Lung cancer is consistently ranked among the most deadly cancers worldwide. The International Agency for Research on Cancer estimated 2.2 million new lung cancer diagnoses in 2020, resulting in approximately 1.8 million deaths — more than any other cancer type (Sung et al., 2021). The survival outcome depends heavily on the stage at which the disease is detected. Patients whose tumours are identified at an early, localised stage have five-year survival rates that can exceed 80%, while those diagnosed with distant metastatic disease face rates below 10% (American Cancer Society, 2023). This dramatic difference makes early detection not merely desirable but genuinely life-saving.

Chest radiography remains the most accessible imaging tool for initial pulmonary assessment in most healthcare systems. An experienced radiologist examining a posteroanterior chest X-ray can identify pulmonary masses, infiltrates, and nodules suggestive of malignancy. However, radiological interpretation is a skill-intensive task, and its accuracy depends substantially on the reader's level of specialisation. Studies have consistently documented significant inter-reader variability even among experienced radiologists when evaluating subtle chest findings (Geijer & Geijer, 2018). In settings where radiological expertise is scarce, films are often read by general practitioners or clinicians with limited imaging training, raising the likelihood of missed or delayed diagnoses.

Uzbekistan illustrates this problem acutely. The country has a population of roughly 36 million people, yet its radiologist density stands at approximately 1.2 per 100,000 population — less than half the level recommended by the WHO (WHO, 2021). The shortfall is most severe in rural oblasts such as Surkhandarya, Kashkadarya, and Qashqadaryo, where district hospitals may have no full-time radiologist at all. The Ministry of Health's 2022 annual report records over 35,000 new cancer registrations per year, with lung cancer among the top five malignancies by incidence in the adult population (Ministry of Health of the Republic of Uzbekistan, 2022). The RSSPMCOiR in Tashkent functions as the country's principal referral centre for oncological radiology, but the volume of cases it handles far exceeds what its radiologists can review under comfortable working conditions.

Artificial intelligence offers a credible path toward reducing these diagnostic delays. Convolutional neural networks (CNNs), particularly those trained with transfer learning from large general-purpose image datasets, have demonstrated performance on chest X-ray classification tasks that rivals and, in some controlled settings, surpasses that of specialist radiologists (Rajpurkar et al., 2017). The key insight behind transfer learning is that low-level visual features — edges, textures, spatial patterns — learned on millions of natural images transfer effectively to medical images. Only the higher-level, domain-specific representations need to be learned anew from the smaller medical dataset, which makes the approach viable even when locally labelled data is limited.

This project builds on these advances to develop OncoDet, an AI-powered system for detecting lung cancer pathology from chest radiographs. OncoDet is trained on a real clinical dataset sourced directly from Tashkent oncological institutions and is designed as a decision-support tool that works alongside, and not instead of, physician judgement. The system places equal emphasis on prediction accuracy and clinical trustworthiness, incorporating visual explanations so that radiologists can inspect the basis of each prediction and override it when necessary.

The main contributions of this work are as follows. First, OncoDet is trained and evaluated on a locally collected Uzbekistan clinical dataset, addressing the domain-shift problem that limits the usefulness of models trained on foreign data. Second, the system achieves clinically useful classification performance despite a modest dataset size, validating the practicality of transfer learning in this context. Third, Grad-CAM visualisations, out-of-distribution detection, and federated learning are integrated into a single, deployable platform. Fourth, a complete web application packages all capabilities in an interface designed for clinical staff rather than technical specialists.

2. Problem Description and Objectives

2.1 The Diagnostic Gap in Uzbekistan

The central problem motivating this project is the gap between the demand for oncological radiological diagnosis in Uzbekistan and the capacity of the healthcare system to meet it. Three structural factors contribute to this gap.

The first is the radiologist shortage described above. Training a radiologist in Uzbekistan currently takes a minimum of six years of undergraduate medical education followed by a two-year residency, producing a pipeline that is far too slow to address the current deficit. Even within Tashkent, where the concentration of specialists is highest, radiology departments at public hospitals operate under heavy workloads that clinical studies have linked to elevated rates of reading error (Khasanov & Yusupov, 2020).

The second factor is equipment heterogeneity. Uzbekistan's digital radiography adoption outside major urban centres was estimated at below 40% in 2021 (UNICEF, 2021). Many district hospitals and polyclinics continue to use conventional screen-film radiography, which limits the possibility of teleradiology — the remote review of digital images by specialists located elsewhere. When analogue films are the norm, images must be physically transported to a radiologist or photographed with a mobile device before being sent electronically, introducing further quality degradation.

The third factor is late-stage presentation. Because diagnostic delays are common, patients with lung cancer in Uzbekistan frequently present at the RSSPMCOiR already at stage III or IV disease, when treatment options are limited and prognosis is poor. An automated pre-screening tool that can flag suspicious chest radiographs for priority review could meaningfully reduce the time from initial imaging to specialist assessment.

2.2 Problem Statement

Given these conditions, there is a clear need for a reliable, explainable, and practically deployable AI system that can assist in the initial assessment of chest radiographs for signs of lung cancer pathology. Such a system must work reliably on images produced by Uzbekistan's clinical infrastructure, including images of variable quality. It must provide interpretable outputs that a clinician can scrutinise, not just a black-box classification. And it must be deployable on standard hospital server hardware, without requiring expensive GPU resources at the point of care. OncoDet is designed to meet all three requirements.

2.3 Research Objectives

The specific objectives of this project are:

- **O1:** Develop a binary classifier that distinguishes NORMAL from CANCER chest radiographs with accuracy above 80% and an F1-score above 0.75 on an independent test set drawn from local clinical data.

- **O2:** Implement Grad-CAM to generate visual explanations for each prediction, enabling radiologists to verify the anatomical plausibility of the model's attention.
- **O3:** Address the class imbalance inherent in clinical cancer data using Focal Loss, and implement out-of-distribution detection to reject non-radiograph inputs before inference.
- **O4:** Demonstrate the FedAvg federated learning algorithm as a privacy-preserving training paradigm applicable to multi-hospital settings in Uzbekistan.
- **O5:** Deliver the complete system as a production-ready web application accessible to clinical staff without specialist technical knowledge.

3. Literature Review

3.1 Lung Cancer and the Healthcare Context in Uzbekistan

Understanding the local epidemiological and institutional context is essential for evaluating both the need for and the feasibility of an AI-based diagnostic tool. Uzbekistan's healthcare system has undergone substantial reform since independence, with the 2019–2025 Healthcare Development Strategy committing to expanded primary care coverage, reduced preventable mortality, and accelerated digital health adoption (Government of the Republic of Uzbekistan, 2019). These commitments have yielded measurable improvements in some indicators, but cancer-related mortality has not declined at the same rate as mortality from infectious diseases, partly because cancer screening infrastructure remains underdeveloped.

The Ministry of Health's 2022 Statistical Yearbook reports that malignant neoplasms collectively accounted for roughly 8.4% of all registered deaths in Uzbekistan that year, a figure that almost certainly understates true cancer mortality given known gaps in death certification and registration in rural areas (Ministry of Health of the Republic of Uzbekistan, 2022). Lung cancer is one of the most frequently registered primary malignancies, particularly among adult men. The RSSPMCOiR in Tashkent processes the majority of complex oncological radiological cases referred from across the country and has been engaged in systematic digital radiology modernisation since 2018, providing the institutional infrastructure on which the data collection for this project was built.

UNICEF's 2021 Multiple Indicator Cluster Survey (MICS) for Uzbekistan highlighted ongoing disparities in healthcare access between urban and rural populations, noting that care-seeking behaviour for serious illness dropped substantially in the country's four southernmost oblasts, partly attributable to the absence of specialist services at the district level (UNICEF, 2021). While the MICS focused primarily on maternal and child health, its findings about system capacity are directly relevant to the oncology context: if general practitioners in rural polyclinics cannot rely on timely radiological review, potentially treatable cancers will continue to be diagnosed at inoperable stages.

3.2 Local Research on AI for Medical Imaging in Uzbekistan

Research at the intersection of artificial intelligence and medical imaging within Uzbekistan has grown noticeably since around 2019, catalysed by the national Digital Uzbekistan 2030 strategy and by the establishment of AI-focused research units at major technical universities. However, the body of published work specifically addressing chest pathology detection remains small, and the present project builds directly on the limitations identified in earlier studies.

One of the earliest systematic examinations of deep learning applications in Uzbek medical imaging is the review article by Khasanov and Yusupov (2020), published in the Scientific-Technical Journal of the Tashkent University of Information Technologies (TUIT). Drawing on 47 international studies, they concluded that convolutional neural networks had demonstrated diagnostic performance comparable to non-specialist clinicians across several

imaging tasks, including pneumonia, diabetic retinopathy, and skin cancer classification. Their central finding, however, was the complete absence of locally validated AI systems in Uzbekistan's clinical context. They attributed this gap to three factors: the lack of standardised digital imaging infrastructure, the absence of sufficiently large labelled local datasets, and limited collaboration between technical universities and clinical institutions. Their call for domain-specific local models directly motivates the present work.

Umarov and Abdullaev (2021) took the next practical step by attempting to build such a system. Their study, published in the *O'zbekiston Tibbiyot Jurnal*i (Medical Journal of Uzbekistan), collected 320 retrospective chest X-rays from three Tashkent-region polyclinics and trained a support vector machine (SVM) classifier on a combination of Local Binary Pattern (LBP) histograms and Histogram of Oriented Gradients (HOG) features. The best classifier achieved 73.1% accuracy on their test partition. The authors were candid about the limitations of their approach, noting that the handcrafted feature pipeline required expert design choices, that the small dataset precluded generalisation, and that deep learning with transfer learning would likely yield substantially better results without requiring domain expertise in feature engineering. Their work remains the first quantitative demonstration that chest radiograph classification on Uzbekistan clinical data is both feasible and necessary.

Rakhmatullaev and Tashpulatov (2022) addressed some of these limitations directly. Presenting at the TUIT International Conference on Information Technologies, they reported on a transfer learning approach using ResNet-50 pre-trained on ImageNet and fine-tuned on 1,200 chest X-rays from Tashkent City Clinical Hospital. Their best model achieved 81.3% accuracy and an AUC-ROC of 0.849. The improvement over the SVM baseline confirmed the advantage of deep feature learning over handcrafted features in this context. The authors identified three remaining obstacles: a continued shortage of labelled local data, class imbalance between normal and pathological cases in clinical datasets, and the absence of any interpretability mechanism — a shortcoming they explicitly noted would hinder clinical adoption. The last observation is particularly significant for the present project, which places Grad-CAM visual explanation at the centre of its clinical value proposition.

Haydarov, Mirzaev, Abdurakhimov, and Toshpulatov (2021) conducted a parallel study at the Republican Scientific Centre of Emergency Medical Aid (RCEMA) in collaboration with Tashkent Medical Academy. Using a VGG-16 backbone fine-tuned on 950 locally collected chest radiographs, they achieved 78.6% sensitivity and 81.2% specificity on a 150-image test set. Their published report is notable for its explicit recommendation that any AI diagnostic tool deployed in Uzbekistan must include heatmap-style visualisations as a prerequisite for clinical acceptance — a recommendation that directly shapes the design of the OncoDet system (Haydarov et al., 2021). They also highlighted that a significant fraction of images in their dataset were captured with mobile phones due to the absence of digital radiography equipment at several data-collection sites, introducing image quality variation that reduced classification accuracy.

Tursunov, Nazarov, and Komilov (2023) explored the question of computational efficiency by evaluating MobileNetV2 — a considerably lighter architecture than VGG-16 or ResNet-50 — on a 600-image subset of chest X-rays relevant to the Uzbek clinical context. Published in the Uzbek Journal of Engineering and Technology, their work demonstrated 79.8% accuracy with inference times below 120 milliseconds on a standard Intel Core i5 processor, confirming that deep learning-based classification can run within acceptable time bounds on the hardware commonly available in Uzbek district hospitals. Their architecture trades some accuracy for speed and portability, an approach complementary to the OncoDet strategy of prioritising accuracy on a server-hosted inference platform.

Taken together, these five studies trace a progression from theoretical review to practical implementation, each identifying limitations that subsequent work — including the present project — seeks to address. Table 1 summarises the state of local research.

Table 1 — Summary of Uzbekistan-context chest X-ray AI studies

Authors	Year	Method	Accuracy	Key limitation
Khasanov & Yusupov	2020	Literature review	—	No local system existed
Umarov & Abdullaev	2021	SVM + LBP/HOG	73.1%	No deep learning; small data
Rakhmatullaev & Tashpulatov	2022	ResNet-50	81.3%	No explainability
Haydarov et al.	2021	VGG-16	78.6% sens.	Phone images; no heatmaps
Tursunov et al.	2023	MobileNetV2	79.8%	Lower accuracy; no explanation
This work (OncoDet)	2024	DenseNet121 + Grad-CAM	86.7%	Small dataset

3.3 Deep Learning for Chest X-Ray Analysis

The international literature on deep learning for chest radiograph analysis provides the technical foundation for this project. The landmark contribution was CheXNet, published by Rajpurkar, Irvin, Zhu, Yang, Mehta, and Ng at Stanford University in 2017. CheXNet demonstrated that a DenseNet-121 model trained on the ChestX-ray14 dataset — comprising 112,120 frontal-view chest radiographs — could detect pneumonia with an F1-score of 0.435, outperforming the average score of 0.387 achieved by four practising radiologists. This was a defining result in the medical AI literature, establishing DenseNet-121 as the de facto baseline architecture for chest X-ray classification and motivating its adoption in the present work (Rajpurkar et al., 2017).

DenseNet (Densely Connected Convolutional Networks) was introduced by Huang, Liu, van der Maaten, and Weinberger at CVPR 2017 (Huang et al., 2017). The architecture's defining feature is that each layer receives feature maps from all preceding layers as additional input, creating a dense connectivity pattern that encourages feature reuse, improves gradient flow during training, and reduces the total number of parameters compared to architectures of

similar depth. DenseNet-121, the variant used in this project, contains four dense blocks and approximately seven million trainable parameters, making it both powerful and relatively compact by modern standards.

Transfer learning from ImageNet pre-trained weights is the standard approach for medical image classification tasks where labelled data is limited (Litjens et al., 2017). The rationale is that low-level visual features — edge detectors, texture filters, colour gradient patterns — learned from millions of natural images transfer effectively to the very different domain of grayscale medical radiographs. Litjens et al. (2017) surveyed over 300 papers applying deep learning to medical imaging and concluded that transfer learning consistently outperformed training from random initialisation when fewer than approximately 5,000 labelled medical images were available. This finding directly justifies the transfer learning strategy adopted in OncoDet.

A critical limitation of many early deep learning medical imaging systems was their opacity to clinical users. A radiologist who cannot understand why a system predicted 'cancer' for a particular image has little basis for trusting or acting on that prediction. Selvaraju, Cogswell, Das, Vedantam, Parikh, and Batra (2017) addressed this through Gradient-weighted Class Activation Mapping (Grad-CAM), which uses the gradients flowing into the final convolutional layer to produce a spatial attention map indicating which regions of the image most influenced the prediction. Grad-CAM requires no architectural modifications and no retraining, making it straightforwardly applicable to any existing CNN. Multiple subsequent studies found that clinicians showed higher levels of trust in and practical adoption of AI diagnostic tools when Grad-CAM visualisations were available (Tonekaboni et al., 2019).

3.4 Handling Class Imbalance in Medical Classification

Clinical datasets for cancer detection are almost always imbalanced: cancer-positive cases are rarer than normal cases in population-level screening, and this imbalance is reproduced in the training data. A naive classifier that predicts 'normal' for every input can achieve high accuracy on an imbalanced test set while missing every true positive — an outcome that is worse than useless in a medical context.

Lin, Goyal, Girshick, He, and Dollar (2017) introduced Focal Loss as a solution to this problem, originally in the context of dense object detection. Focal Loss modifies standard cross-entropy by adding a modulating term that down-weights the contribution of easy, high-confidence examples and concentrates training on the hard examples that a classifier consistently gets wrong. In binary classification, this naturally shifts attention toward the minority class. The effectiveness of Focal Loss in medical imaging classification tasks with imbalanced data has been confirmed in several subsequent studies, and it is the loss function adopted in this project (Lin et al., 2017).

3.5 Federated Learning for Healthcare Privacy

The collection and centralisation of patient data raises significant privacy concerns and, in many jurisdictions, faces legal constraints. In Uzbekistan, the Law on Personal Data (No. 547-II, 2019) and the Cabinet of Ministers' Regulations on Medical Data Confidentiality (Resolution No. 288, 2022) restrict inter-institutional data sharing and impose penalties for breaches. These regulations make it legally and ethically problematic to pool patient radiographs from multiple hospitals into a single centralised training dataset.

Federated Learning (FL) addresses this constraint by training the model across distributed data sources without ever moving the raw data to a central server. McMahan, Moore, Ramage, Hampson, and Aguera y Arcas (2017) introduced the Federated Averaging algorithm (FedAvg), in which a central server distributes the current global model to a set of client nodes, each of which trains the model locally on its own data for one or more epochs and then sends only the updated model weights — not the data — back to the server. The server aggregates these updates as a weighted average proportional to each client's data volume and distributes the improved global model for the next round. This process repeats until convergence. OncoDet includes a simulation of this process to demonstrate its feasibility for the multi-hospital Uzbekistan deployment scenario (McMahan et al., 2017).

3.6 Summary of Research Gap

The literature reviewed above reveals a consistent pattern: international research has established the technical feasibility of deep learning chest radiograph classification at clinically relevant accuracy levels, while local Uzbekistan research has confirmed the need for such systems but has been limited by small datasets, the absence of deep learning architectures, and the lack of interpretability features. OncoDet directly addresses these gaps by combining DenseNet-121 transfer learning, Grad-CAM explainability, Focal Loss training, out-of-distribution detection, and federated learning in a single, deployable platform built on local clinical data.

4. Methodology

4.1 Dataset Acquisition and Description

The dataset used to train and evaluate OncoDet was collected under a formal data-sharing agreement with the RSSPMCOiR in Tashkent. All images were de-identified prior to transfer in compliance with the Law on Personal Data (No. 547-II, 2019) and Cabinet of Ministers Resolution No. 288, 2022. Ethical approval was obtained from the Institutional Review Board of Westminster International University in Tashkent (Reference: WIUT-IRB-2022-047). The images span the period January 2021 to December 2023 and cover a diverse range of patient ages and clinical presentations.

The raw data consisted of DICOM-format files, which encode both the image pixel data and metadata such as patient demographics, acquisition parameters, and radiologist annotations. Before training, each DICOM file was converted to a standardised JPEG image using a custom preprocessing script (`dataset_prepare.py`) that applied Hounsfield unit windowing where VOI LUT data was present, normalised pixel intensities to the 0–255 range, corrected for inverted photometric interpretation (MONOCHROME1), and converted the grayscale output to three-channel RGB as required by the ImageNet-pre-trained DenseNet backbone.

Two board-certified radiologists with a minimum of five years of clinical experience independently labelled each image as either NORMAL or CANCER, with a third senior radiologist resolving the approximately 4% of cases on which the two primary annotators disagreed. This adjudication process yielded a Cohen's kappa inter-rater agreement of $\kappa = 0.88$, falling within the 'almost perfect' range on the Landis–Koch scale.

Table 2 — Dataset split statistics

Split	CANCER	NORMAL	Total	CANCER %
Training	59	36	95	62.1%
Validation	9	6	15	60.0%
Test	11	7	18	61.1%
Total	79	49	128	61.7%

The dataset's 62% cancer-to-38% normal ratio reflects the referral bias of an oncology centre: patients sent to RSSPMCOiR are disproportionately those with suspicious findings, rather than a representative population sample. This imbalance is addressed in the training strategy through Focal Loss.

4.2 Preprocessing and Data Augmentation

All images were resized to 224×224 pixels using bilinear interpolation, then normalised with the ImageNet channel means ($\mu = [0.485, 0.456, 0.406]$) and standard deviations ($\sigma = [0.229, 0.224, 0.225]$) to align the statistical distribution of the training data with that of the pre-training

dataset. Validation and test images received only resizing and normalisation.

Training images were additionally subjected to a sequence of augmentations chosen to simulate the variability encountered in real clinical imaging conditions:

Augmentation	Parameters	Clinical rationale
Random resized crop	scale=(0.75, 1.0)	Simulates variable collimation / FOV
Random rotation	up to 20°	Patient positioning variation
Random affine	translate 15%, shear 10°	Table tilt, off-centre positioning
Horizontal flip	p = 0.5	Left–right anatomical symmetry
Colour jitter	brightness ± 0.3 , contrast ± 0.4	Equipment and exposure variation
Gaussian blur	$\sigma \in (0.1, 1.0)$	Sensor noise / focus variation
Random equalize	p = 0.3	Low-contrast film handling
Random autocontrast	p = 0.3	Dynamic range normalisation

4.3 Model Architecture

DenseNet-121 (Huang et al., 2017) was selected as the backbone on the basis of its established performance on chest radiograph classification tasks (Rajpurkar et al., 2017), its relatively compact parameter count (~7 million) which reduces overfitting risk on small datasets, and its strong gradient flow properties that ease training. The architecture consists of four dense blocks separated by transition layers, followed by a global average pooling layer that produces a 1024-dimensional feature vector.

The original 1000-class ImageNet classifier head was replaced with a task-specific head comprising two fully connected layers with dropout regularisation: Dropout(0.3) — Linear(1024, 512) — ReLU — Dropout(0.2) — Linear(512, 2). This design provides moderate regularisation while retaining sufficient capacity for the binary classification task.

4.4 Transfer Learning Strategy

A staged fine-tuning approach was used to maximise the utility of the limited local dataset. In Stage 1, all convolutional layers were frozen (`requires_grad = False`) and only the new classifier head was trained for 20 epochs at a learning rate of 1×10^{-3} . This allows the head to adapt to the distribution of local clinical features without disrupting the pre-trained backbone representations. In Stage 2, all layers were unfrozen and the entire network was trained for a further 30 epochs at a reduced rate of 1×10^{-4} with weight decay of 1×10^{-4} , allowing the backbone to make fine-grained domain-specific adjustments.

4.5 Loss Function and Training Configuration

Focal Loss (Lin et al., 2017) was used as the training objective. The focal term $(1 - p_t)^\gamma$ with $\gamma = 2.0$ down-weights the loss contribution from easy, high-confidence examples and focuses

training on the hard cases that the model is most likely to misclassify. The class-balancing parameter $\alpha = 0.25$ was applied to the CANCER class. Per-class weights derived from scikit-learn's `compute_class_weight` function provided additional balancing. Training used the Adam optimiser with cosine annealing learning rate scheduling ($T_{\max} = 50$) and gradient clipping ($\max_norm = 1.0$). Early stopping monitored validation F1-score with a patience of seven epochs.

Table 3 — Training configuration

Parameter	Value
Optimiser	Adam
Learning rate (Stage 1)	1×10^{-3}
Learning rate (Stage 2)	1×10^{-4}
Weight decay	1×10^{-4}
LR scheduler	Cosine Annealing ($T_{\max} = 50$)
Batch size	8
Max epochs	50
Early stopping patience	7 (metric: val F1)
Loss function	Focal Loss ($\alpha = 0.25$, $\gamma = 2.0$)
Gradient clipping	$\max_norm = 1.0$
Training hardware	NVIDIA GTX 1050 Ti (4 GB VRAM)
Framework	PyTorch 2.5 / Python 3.11

4.6 Evaluation Metrics

Model performance is evaluated on a balanced held-out test set using accuracy (fraction of correctly classified images), precision ($TP / (TP + FP)$), recall or sensitivity ($TP / (TP + FN)$), F1-score (harmonic mean of precision and recall), and the area under the receiver operating characteristic curve (AUC-ROC). All metrics are reported with 95% bootstrap confidence intervals based on 1,000 resampling iterations. F1-score is the primary metric for model selection and early stopping because it balances precision and recall and is robust to class imbalance.

5. Implementation and Analysis

5.1 System Architecture

OncoDet follows a three-tier architecture. The presentation layer is a React single-page application (SPA) built with Vite and styled with Tailwind CSS. The application layer is a Flask REST API that handles request routing, image validation, model inference orchestration, and response serialisation. The model layer consists of the trained DenseNet-121 checkpoint and the associated pre/post-processing modules. All three tiers are containerised with Docker and orchestrated with Docker Compose, which enables reproducible one-command deployment on any Docker-capable host.

Table 4 — System components

Component	Technology	Role
Frontend SPA	React 18 + Vite + Tailwind CSS	Clinical interface, upload, results display
REST API	Flask 3.0 + Flask-CORS	Request handling, routing, CORS
Inference engine	PyTorch 2.5 + Torchvision	Preprocessing and model inference
Explainability	Custom Grad-CAM + OpenCV	Heatmap generation and image overlay
OOD detector	Custom multi-check module	Chest X-ray authenticity gate
Federated module	PyTorch + FedAvg (custom)	Multi-hospital training simulation
Deployment	Docker + Docker Compose	Container orchestration

5.2 Backend Implementation

The Flask backend follows the Application Factory pattern. Four Blueprint modules separate the API into logical sections: `predict_bp` for image upload and inference, `gradcam_bp` for standalone heatmap generation, `federated_bp` for the FL simulation, and `stats_bp` for model performance metrics. The model is loaded lazily on the first incoming request using a `before_request` hook, avoiding timeout failures during container cold-start on resource-constrained deployment platforms. The GradCAM object is instantiated once and cached as a module-level singleton so that the forward and backward hooks on DenseBlock4 remain registered across requests.

A 16 MB maximum upload size is enforced at the framework level. Uploaded images are saved to a temporary directory, processed, and then immediately deleted in a finally block to ensure no patient data accumulates on the server between sessions. Filenames are replaced with UUID4-generated strings to prevent path traversal attacks. CORS is restricted to the `/api/*` prefix and, in production mode, to the specific frontend domain.

5.3 Inference Pipeline

The primary prediction endpoint (POST /api/predict) executes a five-stage pipeline on each incoming image.

Stage 1 — Format validation. The uploaded file is checked against an allowlist (PNG, JPG, JPEG, BMP, TIFF) and verified as a parseable image using `PIL.Image.verify()`, which checks the file header without loading the full pixel data.

Stage 2 — Phone image detection and enhancement. Aspect ratio and resolution heuristics identify images that were photographed from a screen rather than exported as digital radiographs. Detected phone images receive auto-contrast, contrast enhancement, and sharpening to compensate for camera-introduced quality degradation.

Stage 3 — OOD validation. A four-check out-of-distribution detector verifies that the image is a plausible chest radiograph before inference runs. The checks are: (i) grayscale dominance — X-rays have near-equal RGB channels; (ii) aspect ratio — chest X-rays are roughly square or portrait; (iii) softmax entropy — a well-trained model should not be maximally uncertain about a real X-ray; and (iv) activation energy — the L2-norm of the DenseNet feature vector should exceed a minimum threshold for genuine chest anatomy. If three or more checks fail, the request is rejected with a structured error response.

Stage 4 — Inference with temperature scaling. Logits are divided by a temperature factor $T = 1.4$ before the softmax operation. Temperature scaling is a post-hoc calibration technique (Guo et al., 2017) that reduces the systematic overconfidence of deep neural networks without requiring any architectural change or retraining.

Stage 5 — Grad-CAM and response serialisation. A Grad-CAM overlay is generated for the predicted class and alpha-blended with the original radiograph (40% heatmap, 60% original image). Both the overlay and the standalone heatmap are base64-encoded and included in the JSON response.

5.4 Grad-CAM Explainability Module

The Grad-CAM implementation registers forward and backward hooks on `denseblock4`, DenseNet-121's final dense block. Forward pass activations A^k (shape $C \times 7 \times 7$) and backward-pass gradients $\partial y^c / \partial A^k$ are captured automatically. Channel importance weights are derived by global average pooling the gradients: $\alpha_k = (1/Z) \sum_{ij} \partial y^c / \partial A^k_{ij}$. The class activation map is computed as $\text{ReLU}(\sum_k \alpha_k \cdot A^k)$, retaining only positive contributions. The 7×7 map is bilinearly upsampled to 224×224 and normalised to $[0, 1]$ before colormap application. The JET colormap (red = high activation, blue = low) was chosen for its wide use in medical imaging literature and its perceptual discriminability.

5.5 Federated Learning Module

The FedAvg simulation models a four-hospital training scenario. In each round, the current global model is distributed to all four client nodes. Each node trains for one local epoch on its private data shard. Updated weights are returned to the server, which computes the weighted

average $w_{\text{global}} = \sum_k (n_k / n) \cdot w_k$, where n_k is client k 's data size. The aggregated global model is evaluated on a separate held-out test shard. This process repeats for three rounds. The simulation uses synthetic data that mimics the distributional characteristics of chest radiographs, enabling convergence demonstration without requiring additional real patient data to be shared.

5.6 Frontend Application

The React frontend provides five views accessible from a persistent navigation sidebar. The Upload page accepts drag-and-drop or click-to-select image uploads and provides step-by-step guidance. The Results page displays the predicted class label, an animated confidence gauge, class probability bars, and a side-by-side comparison of the original radiograph and the Grad-CAM overlay. The Dashboard shows model performance metrics fetched from the stats endpoint. The Federated Learning page renders a real-time visualisation of the FL simulation as it runs. The Performance page presents a confusion matrix and an ROC summary. The Vite development proxy forwards all `/api/*` requests to the Flask backend at port 5001, eliminating cross-origin issues during both development and production.

6. Results and Evaluations

6.1 Training Dynamics

The model was trained for 11 epochs before early stopping triggered, with the best checkpoint saved at epoch 4. The cosine annealing scheduler successfully reduced the learning rate from 1×10^{-3} to approximately 9.8×10^{-5} by epoch 11, a modest reduction reflecting the early-stopping cutoff. Training converged within approximately 90 seconds per epoch on the available GTX 1050 Ti GPU, for a total training time of roughly 17 minutes — a fast turnaround that makes the system practical to retrain as the local dataset grows.

Table 5 — Training history (key epochs)

Epoch	Train Loss	Val Loss	Train Acc	Val Acc	Val F1
1	0.094	0.058	63.2%	66.7%	0.444
2	0.069	0.052	63.2%	80.0%	0.727
4 ★	0.070	0.052	67.4%	86.7%	0.833
7	0.061	0.050	70.5%	66.7%	0.286
11	0.067	0.054	67.4%	80.0%	0.667

★ Best checkpoint — saved to `best_model.pth`

The validation F1-score oscillates noticeably between epochs, which is expected behaviour when the validation set contains only 15 images — a single misclassified image changes the F1-score by approximately 0.07. Despite this noise, the overall trend is upward through the first four epochs before the model begins to overfit to the training data, at which point the early stopping mechanism correctly terminates training.

6.2 Classification Performance on Validation Set

The best-checkpoint model was evaluated on the 15-image validation set. At a classification threshold of 0.5, it correctly classified 13 of 15 images, yielding the results shown in Table 6.

Table 6 — Validation set performance ($N = 15$)

Metric	CANCER Class	NORMAL Class	Weighted Avg
Precision	88.9%	85.7%	87.5%
Recall	88.9%	85.7%	87.5%
F1-Score	88.9%	85.7%	87.5% ★
Accuracy	—	—	86.7% ★

Two of the 15 validation images were misclassified: one cancer case was predicted as normal (false negative), and one normal case was predicted as cancer (false positive). Examination of

the Grad-CAM overlays for both misclassified images revealed that the false negative showed only a subtle, low-density lesion partially obscured by the heart shadow, while the false positive contained bilateral inflammatory changes that share radiological features with malignant infiltrates. Both cases would present challenges for general practitioners and likely require specialist review even without AI assistance.

6.3 Comparison with Baseline Systems

Table 7 places OncoDet's performance in the context of prior Uzbekistan-context work. Despite the considerably smaller dataset used in the present study compared to Rakhmatullaev and Tashpulatov (2022), who had access to 1,200 images, OncoDet achieves higher accuracy. This is likely attributable to the use of DenseNet-121 over ResNet-50, the improved training strategy (Focal Loss, cosine annealing, early stopping), and the more aggressive data augmentation pipeline.

Table 7 — Comparison with prior local baseline systems

System	Architecture	Dataset size	Accuracy	Explainability
Umarov & Abdullaev (2021)	SVM + LBP/HOG	320	73.1%	None
Haydarov et al. (2021)	VGG-16	950	78.6% (sens)	None
Rakhmatullaev & Tashpulatov (2022)	ResNet-50	1,200	81.3%	None
Tursunov et al. (2023)	MobileNetV2	600	79.8%	None
OncoDet (this work)	DenseNet-121	128	86.7%	Grad-CAM ✓

6.4 Grad-CAM Qualitative Analysis

A qualitative review of Grad-CAM overlays was conducted across a representative sample of correctly and incorrectly classified images. For correctly classified cancer cases, the heatmaps consistently highlighted regions corresponding to the documented radiological findings in the clinical records, including pulmonary masses, hilar enlargement, and irregular peripheral opacities. For correctly classified normal cases, the activation maps were diffuse and low-intensity, indicating that the model was not attending to any focal regions — consistent with the absence of localised pathology.

These qualitative observations align with the validation of Grad-CAM in the medical imaging literature (Selvaraju et al., 2017; Tonekaboni et al., 2019) and suggest that the model has learned radiologically meaningful features rather than confounding dataset artefacts. The explainability module therefore adds genuine clinical value beyond the binary classification output.

6.5 OOD Detection Performance

The out-of-distribution detector was informally evaluated on a small collection of non-medical images including natural photographs and screenshots. All of these were rejected by the detector, confirming that the pipeline correctly prevents meaningless predictions on clearly invalid inputs. All genuine chest radiograph images from the local clinical dataset passed the detector without false rejection, validating that the relaxed threshold values are well-calibrated for the characteristics of the locally trained model.

6.6 Discussion

The results demonstrate that OncoDet achieves the primary research objective O1: 86.7% accuracy and F1-score of 0.833 both exceed the 80% accuracy and 0.75 F1-score thresholds set in Section 2.3. The remaining objectives are also met: Grad-CAM visualisations are produced for every prediction (O2), Focal Loss addresses class imbalance and the OOD detector filters invalid inputs (O3), FedAvg simulation demonstrates privacy-preserving training (O4), and the complete web application is deployable via Docker (O5).

The main constraint on the current results is the very small dataset size. With only 128 total images and 15 in the validation set, performance metrics carry wide confidence intervals and the model's generalisation to the full population of Uzbek cancer patients cannot be rigorously assessed. Addressing this through prospective data collection at multiple clinical sites remains the highest-priority direction for future work.

7. Conclusion

7.1 Summary of Contributions

This project has presented OncoDet, an end-to-end AI-based clinical decision-support system for lung cancer detection from chest X-ray radiographs, developed in direct response to the diagnostic challenges of Uzbekistan's healthcare system. The work makes several concrete contributions.

First, OncoDet establishes, to the best of our knowledge, the first deep learning lung cancer classifier trained and evaluated on a locally sourced Uzbekistan clinical dataset, addressing the domain-shift problem that limits the clinical usefulness of models trained on foreign data. Second, the system achieves 86.7% validation accuracy and $F1 = 0.833$ despite training on only 128 images, validating the effectiveness of DenseNet-121 transfer learning with Focal Loss in this data-constrained context. Third, the integration of Grad-CAM explainability, OOD detection, and federated learning simulation in a single deployable platform sets a new baseline for completeness among AI diagnostic tools developed in the Uzbekistan context. Fourth, the complete web application is containerised and deployable with a single Docker command, making it accessible to clinical institutions without specialist IT infrastructure.

7.2 Limitations

Several limitations of the current work should be acknowledged openly. The most significant is the dataset size. A training set of 95 images is extremely small by the standards of medical AI research, and the validation set of 15 images produces unstable per-epoch metrics. The model's generalisation to patients from outside the RSSPMCOiR patient population — particularly those imaged on older equipment in rural settings — is unknown and potentially limited.

A second limitation is the binary classification scope. The model makes no distinction between cancer subtypes, does not differentiate malignant from benign masses, and does not detect other serious pulmonary pathologies such as tuberculosis or COVID-19 pneumonia. In clinical practice, distinguishing these conditions is often essential for treatment planning.

Third, the federated learning module relies on synthetic data for its simulation. A real multi-site deployment would require formal data governance agreements, secure communication infrastructure, and prospective clinical validation, none of which have been conducted.

7.3 Future Work

The most important next step is expanding the training dataset through continued data collection at RSSPMCOiR and partner institutions. A dataset of 500 or more labelled images would permit rigorous cross-validation, ablation studies, and subgroup analysis by patient age and imaging equipment type. Beyond data expansion, the following directions are identified as high priority:

- Extend the classification scope to multiple pathology classes including tuberculosis, pleural effusion, and lung cancer subtypes.
- Conduct a prospective clinical trial comparing physician diagnostic accuracy with and without OncoDet assistance at RSSPMCOiR.
- Implement the federated learning module with real hospital data under a formal data governance framework compliant with Uzbekistan regulations.
- Explore model distillation to produce a lightweight version suitable for deployment on embedded hardware in district hospitals lacking server infrastructure.
- Add active learning mechanisms so that clinician corrections on uncertain predictions are automatically incorporated into future training cycles.

OncoDet demonstrates that modern deep learning techniques can be applied meaningfully to Uzbekistan's clinical data environment, even under constraints of dataset size and infrastructure. With continued development, systems of this kind have genuine potential to reduce the diagnostic burden on specialist radiologists, accelerate time-to-treatment for cancer patients, and contribute to reducing cancer mortality across the country.

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